

RF Emitter Localization with Position-Adaptive MAV Platforms

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Abstract — This paper provides a discussion and a summary of the latest results on a novel sensor-node based approach to emitter localization denoted as Position - Adaptive Direction Finding (PADF). PADF is based on the formulation and investigation of iterative path-loss based (i.e. path loss exponent) metrics estimates that are measured across multiple platforms in order to robotically/intelligently positionally adapt (i.e. self-adjust) the location of each platform. We demonstrate that this approach shows potential for accurate emitter localization in challenging embedded multipath environments (i.e. urban environments).

Index Terms — Position-Adaptive Direction Finding (PADF), RF Emitter Localization, Micro Air Vehicles, Wireless Sensor Node Networks

I. INTRODUCTION

This paper provides a discussion and a summary of the latest results on a novel sensor-node based approach to emitter localization denoted as Position -Adaptive Direction Finding (PADF) [1]. PADF is based on the formulation and investigation of path-loss based (i.e. path loss exponent) metrics that are measured and estimated across multiple platforms in order to robotically/intelligently positionally adapt (i.e. self-adjust) the location of each platform.

We demonstrate that this approach shows potential for accurate emitter localization in challenging embedded multipath environments (i.e. urban environments). Initial experimental test results (with ground-based distributed RF sensor nodes) conducted within the AFRL RF Systems Integration Lab at IDCAST (Institute for Development Commercialization of Advanced Sensor Technology) indicate correlated trends between multi-platform RF-based control metrics formulated for this investigation and localization accuracy. These initial tests are conducted with an emitter inside a partially metal-sealed enclosure where variational hardware simulations are performed across several sets of discrete position-adaptive sensor node configurations/geometries. A concise set of tabulated results that include these correlated parameters are discussed in Section III. Section IV illustrates initial approaches to MAV/Sensor (Micro Air Vehicle) platform

integration and indoor flight test simulation geometries presently under investigation within the AFRL MAV Lab Facility at the AFRL Air Vehicles Directorate.

II. POSITION-ADAPTIVE DIRECTION FINDING CONCEPTS

Fig. 1 illustrates two states of a notional PADF geometry using three UAV's that are integrated with Sensor Motes programmed to function as RF receivers. A

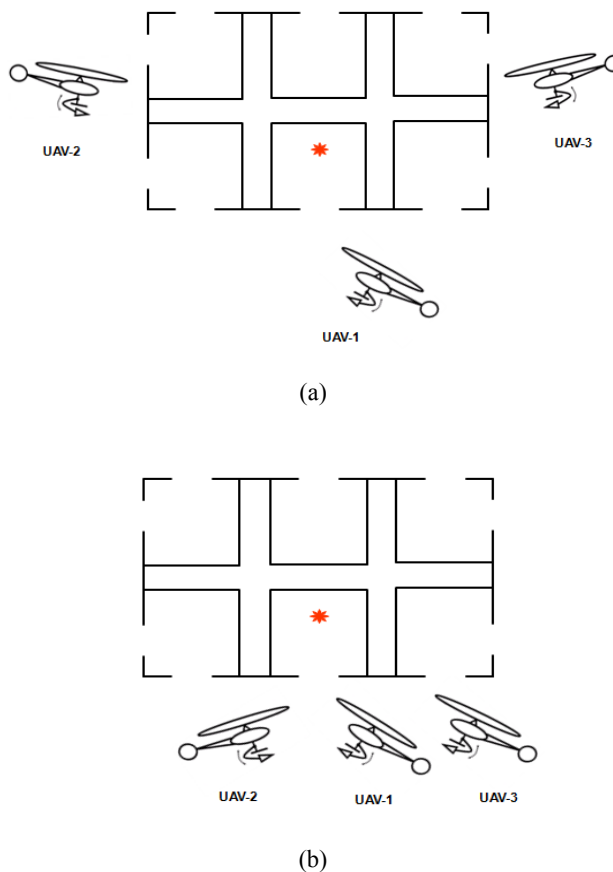


Fig. 1. (a) Notional PADF STATE-0 [1]
(b) Notional PADF STATE-1 (After One Stage Of Multi-Platform Adaptation) [1]

theoretical signal analysis for this geometry is provided in [1] for purposes of formulating and illustrating this technique. As illustrated in Fig. 2, the formulation and investigation of path-loss based (i.e. path loss exponent) metrics that are measured across multiple platforms are intrinsic to this technique. For purposes of preliminary illustration of this concept, a series of approximate sample computations are considered in [1] that correspond to a notional STATE-0 multi-platform geometry as depicted in Fig. 1a and a notional STATE-1 multi-platform geometry as depicted in Fig. 1b. For purposes of illustration, the following path loss exponent based multi-platform control metric for STATE-0, denoted as PLE_{PADF0} was adopted in [1]:

$$PLE_{PADF0} = \min \left(\frac{P_{r1}(dB) - P_{r2}(dB)}{10 \log_{10} \left(\frac{d_2}{d_1} \right)}, \frac{P_{r1}(dB) - P_{r3}(dB)}{10 \log_{10} \left(\frac{d_3}{d_1} \right)} \right) \quad (1)$$

where P_{r1} , P_{r2} , and P_{r3} are based on a theoretical model for RSSI (Receiver Signal Strength Indicator) measurements at UAV Sensor Mote-1, UAV Sensor Mote-2, and UAV Sensor Mote-3, respectively. Similarly, d_1 , d_2 , and d_3 represent theoretical distance estimates from each Sensor Mote to an RF emitter and are also derived from a theoretical model developed for RSSI measurements at each mote.

In this paper, alternative multi-platform control metrics are defined/investigated that estimate PLE values within the closed-loop multi-platform control structure depicted Fig. 2 via a series of cooperative transmissions between the a combination of UAV/UGV platforms. Specifically, the following set of equations, based on RSSI measurements at each mote, are formulated and

implemented for purposes of initiating the investigation of cooperative PLE-based multi-platform control metrics for purposes of localization of non-cooperative emitters:

$$PL_{i,j} = \text{median} \left(10^{\frac{RSSI_{i,j}}{10}} \right) \quad (2)$$

$$PL_{dB_{i,j}} = 10 \log_{10}(PL_{i,j}) \quad (3)$$

$$PLE_{i,j} = \log_{10} \left(10^{\frac{PL_{dB_{i,j}}}{10}} \right) \quad (4)$$

$$\sigma = \text{stdv}(PLE_{i,j}) \quad (5)$$

$$PADF_Metric = \frac{1}{\sigma} \quad (6)$$

where i, j denote cooperative transmissions between Sensor i and Sensor j . The resulting PADF Metric in Eq. 6 is investigated as a multi-platform RF-scattering-based objective function for purposes of multi-platform position adaptation and, in turn, multi-platform localization of embedded emitters.

Localization computations based on distance estimates derived from a set of cooperative RSSI measurements between combinations of four UAV/UGV platforms can be implemented via the tabulation of all combinations of cooperative transmissions between platforms. For this analytical formulation, (x,y) represents a two-dimensional location estimate for an embedded emitter and (x_i, y_i) represents the known locations of each cooperative platform.

Then, the following modified least squares approach can be adopted for this PADF investigation where the distance terms in the A -matrix are weighted by the inverse of vector Cross PLE terms that are estimated from a set of cooperative RSSI measurements between four platforms as described by Eq. 2-6:

$$\overrightarrow{CPL\bar{E}} = [PLE_{1,2} \ PLE_{1,3} \ PLE_{1,4} \ PLE_{2,3} \ PLE_{2,4} \ PLE_{3,4}] \quad (7)$$

$$MPLE = \text{mean}(\overrightarrow{CPL\bar{E}}) \quad (8)$$

$$\overrightarrow{W} = \begin{bmatrix} \frac{MPLE}{PLE_{1,2}} & \frac{MPLE}{PLE_{1,3}} & \frac{MPLE}{PLE_{1,4}} & \frac{MPLE}{PLE_{2,3}} & \frac{MPLE}{PLE_{2,4}} & \frac{MPLE}{PLE_{3,4}} \end{bmatrix} \quad (9)$$

As illustrated by the following derivation, this set of cooperative inverse PLE weights have the relative effect of de-emphasizing measurements that are affected by non-line-of-sight propagation losses due to multipath and materials attenuation.

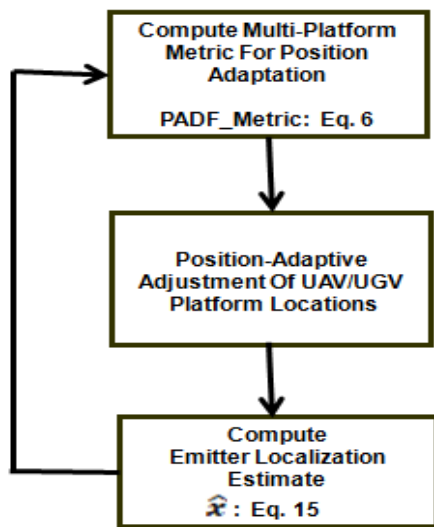


Fig. 2. Basic Functional Block Diagram Model for PADF Development [1]

$$\mathbf{A} = \begin{bmatrix} 2(x_2 - x_1) & 2(y_2 - y_1) \\ 2(x_3 - x_1) & 2(y_3 - y_1) \\ 2(x_4 - x_1) & 2(y_4 - y_1) \\ 2(x_3 - x_2) & 2(y_3 - y_2) \\ 2(x_4 - x_2) & 2(y_4 - y_2) \\ 2(x_4 - x_3) & 2(y_4 - y_3) \end{bmatrix} \quad (10)$$

$$\mathbf{b} = \begin{bmatrix} x_2^2 - x_1^2 + y_2^2 - y_1^2 + d_1^2 - d_2^2 \\ x_3^2 - x_1^2 + y_3^2 - y_1^2 + d_1^2 - d_3^2 \\ x_4^2 - x_1^2 + y_4^2 - y_1^2 + d_1^2 - d_4^2 \\ x_3^2 - x_2^2 + y_3^2 - y_2^2 + d_2^2 - d_3^2 \\ x_4^2 - x_2^2 + y_4^2 - y_2^2 + d_2^2 - d_4^2 \\ x_4^2 - x_3^2 + y_4^2 - y_3^2 + d_3^2 - d_4^2 \end{bmatrix} \quad (11)$$

$$\mathbf{A}_W(i, 1) = \mathbf{A}(i, 1) * \vec{W}(i) \quad (12)$$

$$\mathbf{A}_W(i, 2) = \mathbf{A}(i, 2) * \vec{W}(i) \quad (13)$$

Then, this set of pre-weighted modified least-squares equations along with the two-dimensional localization estimate can be expressed as follows:

$$\mathbf{A}_W \mathbf{x} = \mathbf{b} \quad (14)$$

$$\hat{\mathbf{x}} = (\mathbf{A}_W^T \mathbf{A}_W)^{-1} \mathbf{A}_W^T \mathbf{b} \quad (15)$$

The next two sections provide sets of experimental data analysis using this PADF technique, as depicted in Eqns. 2 – 15, in two different experimental environments.

III. EXPERIMENTS AT IDCAST

During the initial phases of this investigation, a series of basic Sensor Mote tests and distributed software control/measurement functions were developed for purposes of performing PADF experiments and hardware simulations using networked Wireless Sensor Motes [2]. In order to obtain more reliable RSSI measurements, a

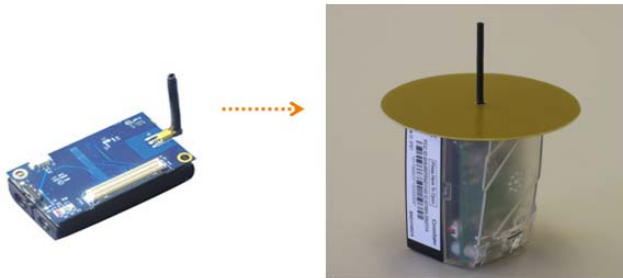


Fig. 3. Customized Motes for Relatively Omni-Directional RSSI Measurements

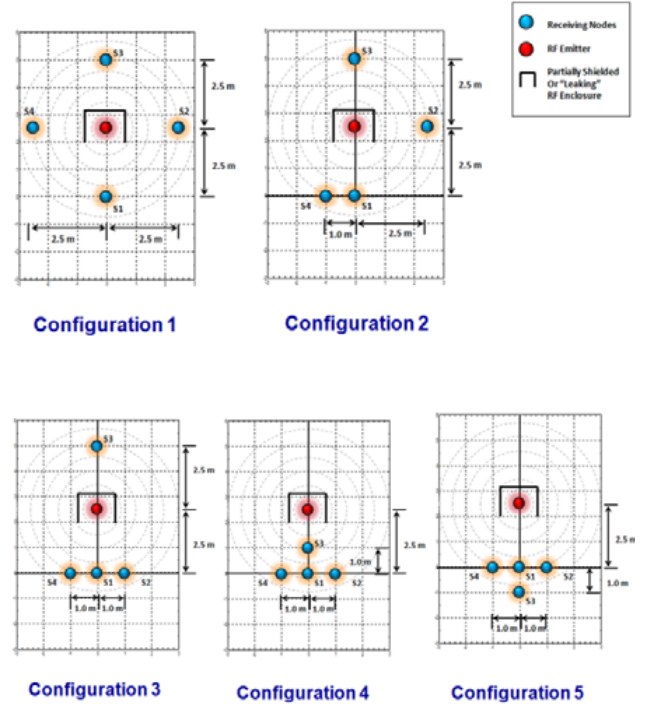


Fig. 4. Discrete Position-Adaptive Configurations for Experiments at IDCAST

portion of this initial experiment design phase was allocated to customizing the aperture plane on the Sensor Motes to as shown in Fig. 3.

After designing and implementing this PADF experimental apparatus with distributed wireless sensor motes, a number of discrete configurations were synthesized for purposes of investigating PADF concepts as outlined in Section II. This set of experimental configurations are shown in Fig. 4. For example, the diagram for Configuration 1 is a top view perspective of a simulated initial, or STATE-0, platform geometry with all four sensor motes spaced 2.5 meters apart in both the x and y dimensions. Similarly, the diagram for Configuration 3 is a top view perspective of a simulated sensor geometry after two iterative position adaptive re-adjustments of the platform locations. In this configuration, three sensor motes are depicted to adapt in position to locations in front of the RF leakage point that is associated with the (aluminum foil covered) enclosure that, in turn, contains the embedded RF emitter in the center. In this case (i.e. Configuration 3), the three sensor motes in front of the enclosure are located 1 meter apart.

Fig. 5 is a collection of pictures that illustrate the experimental apparatus developed for PADF experiments at IDCAST. The partially sealed enclosure with the embedded non-cooperative emitter is shown in the two

pictures on the left and also in the picture on the lower right. The enclosure is covered with aluminum foil on all sides except the front in order to simulate an embedded RF environment with one leakage point. The base station for this wireless sensor network is shown on the upper right.



Fig. 5. Selected Pictures of Experimental Apparatus Developed for PADF Investigations at IDCAST

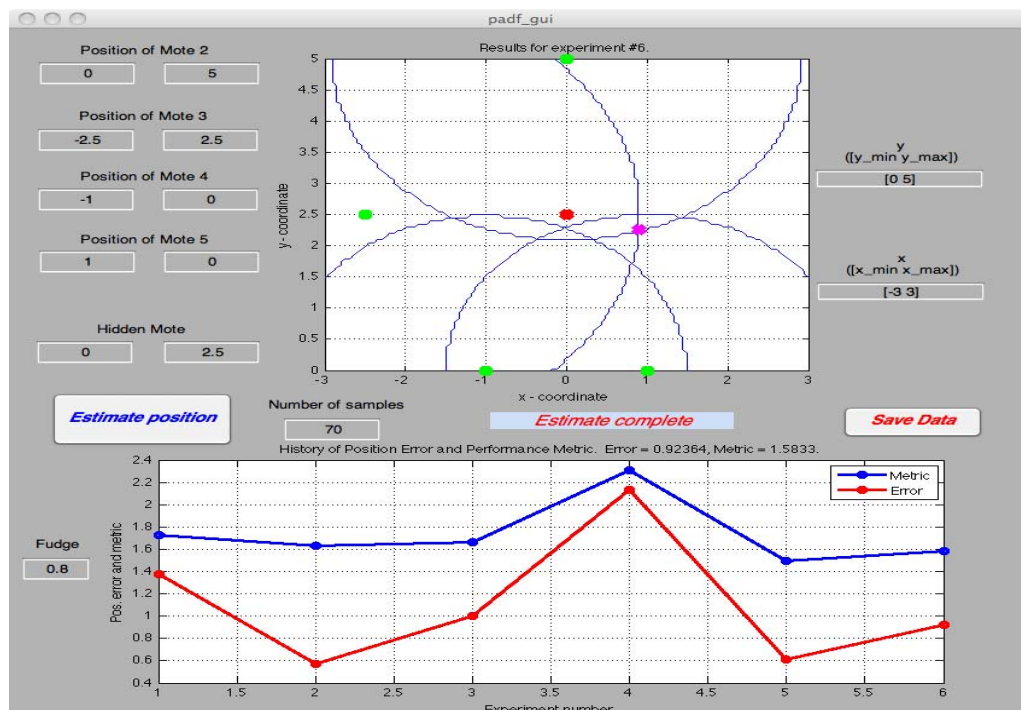


Fig. 6. Selected Pictures of Experimental Apparatus Developed for PADF Investigations at IDCAST

The basic trends and potential candidate approaches to further analysis/development of PADF techniques that are indicated via analysis of this initial IDCAST data set and specifically, the plots contained on the lower portion of the screen capture in Fig. 6 can be observed as follows:

- The Blue Curve is a plot of the PADF Metric (i.e. Eq. 6) versus position-adaptive experiment configuration number (i.e. Configurations 1 – 5 in Fig. 4.)
- The Red Curve is a plot of the Localization Error versus position-adaptive experiment configuration number
- There exists a distinctly observable correlation between the overall trend or shape of the Red Curve and Blue Curve for this particular data set. This type of correlated trend between the two curves implies that the PADF Metric developed during this investigation shows promise for further development and investigation as a multi-platform objective function for purposes of furthering the advancement of PADF systems
- Further perturbation analysis is required in order to gain insight into the possibility of using Position-Adaptive techniques for purposes of potentially “Cherry Picking” select samples from a large data set? This type of capability assessment can be related to the consistency of a given PADF solution in the neighborhood of small localized platform fluctuations. In other words, if PADF-related real-time parameter estimates vary wildly in the neighborhood of localized platform fluctuations, this particular fluctuating region of a PADF performance curve can be denoted as “inconsistent” and not necessarily conducive to potentially obtaining performance gains via the conscious delimitation (or “Cherry Picking”) of data over small localized regions of the multi-platform coverage space. If, on the other hand, a set of PADF-related real-time parameter estimates remain relatively consistent in the neighborhood of localized platform fluctuations, the associated region of the PADF operating space may be relatively appropriate for obtaining performance gains via conscious robotic/intelligent data delimitation or “Cherry Picking” enabled by the Position-Adaptive nature of this technique. This auxiliary approach to obtaining potential performance gains in challenging environments will be illustrated in the next section.
- On a related note, once a multiplatform objective function is shown to converge to a localized optima of a given PADF Metric, is it possible to consider spreading the platforms apart slightly in an effort to increase localization accuracy while maintaining

basic platform locations in the vicinity of a corresponding PADF solution for a given local optima. This approach to increasing localization accuracy can also be mapped to localized consistency measures. In other words, when implementing a procedure to iteratively increase the distance between platforms as a posteriori (post-processing) steps, consistency measures can be investigated that can enable the localization accuracy to increase while maintaining the basic structure of a given localized PADF geometrical optima (i.e. maintaining a PADF solution without complete loss of localization accuracy due to large deviations from the local PADF optima.) This approach to a iterative a posteriori processing and perturbational multi-platform position-adaptations has the effect of restricting the movement of the platforms to within desired limits for good localization accuracy

IV. EXPERIMENTS AT AFRL MAV LAB

As discussed in the last couple of bulletized comments in the previous section, integrating the Networked PADF Mote Sensors from the discretized PADF experiment design at IDCAST onto a combination of continuously adaptable UAV/UGV platforms can enable the investigation of a number of significant refinements to the present baseline PADF technique as described in Section II. These potential refinements are described and modeled within the dotted circle block of the modified functional PADF block diagram in Fig. 7.

In order to further investigate PADF concepts in a indoor MAV Lab environment, a two-mode PADF concept as depicted in Figure 8 is developed for purposes of investigating potential refinements in consistency, sensitivity, and robustness via the design and implementation of three stationary platforms and one continuously moving platform during the MODE2 localization phase.

As depicted in the left half of Fig. 8, in MODE1 a MAV that is integrated with a custom Sensor Mote for Networked Cooperative PADF is flown in a circular trajectory around an experimental region-of-interest. This experimental region-of-interest contains an embedded enclosure that is partially wrapped with aluminum foil and then covered with black cloth to avoid inducing ambiguities in the distributed camera-based optical navigation system. An embedded emitter (i.e. a non-cooperative Sensor Mote) is placed in the embedded closure in the vicinity of a RF leakage point that is cut out from the front of the enclosure. The basic functionality of MODE1 is the detection and demarcation of a contoured sub-interval as indicated by the contour endpoint symbols, C1 and C2, in the MODE1 diagram towards the right half of Fig. 8.

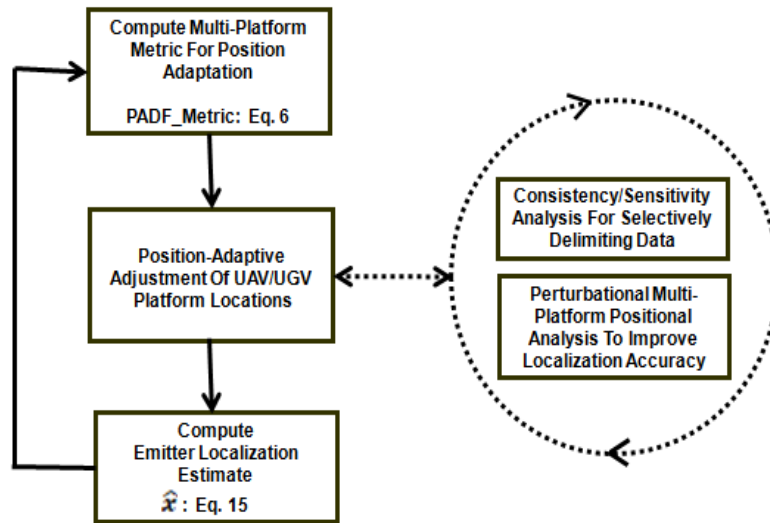


Fig. 7. Functional Block Diagram Model for PADF Development with Optional Steps for Improving Robustness and Accuracy

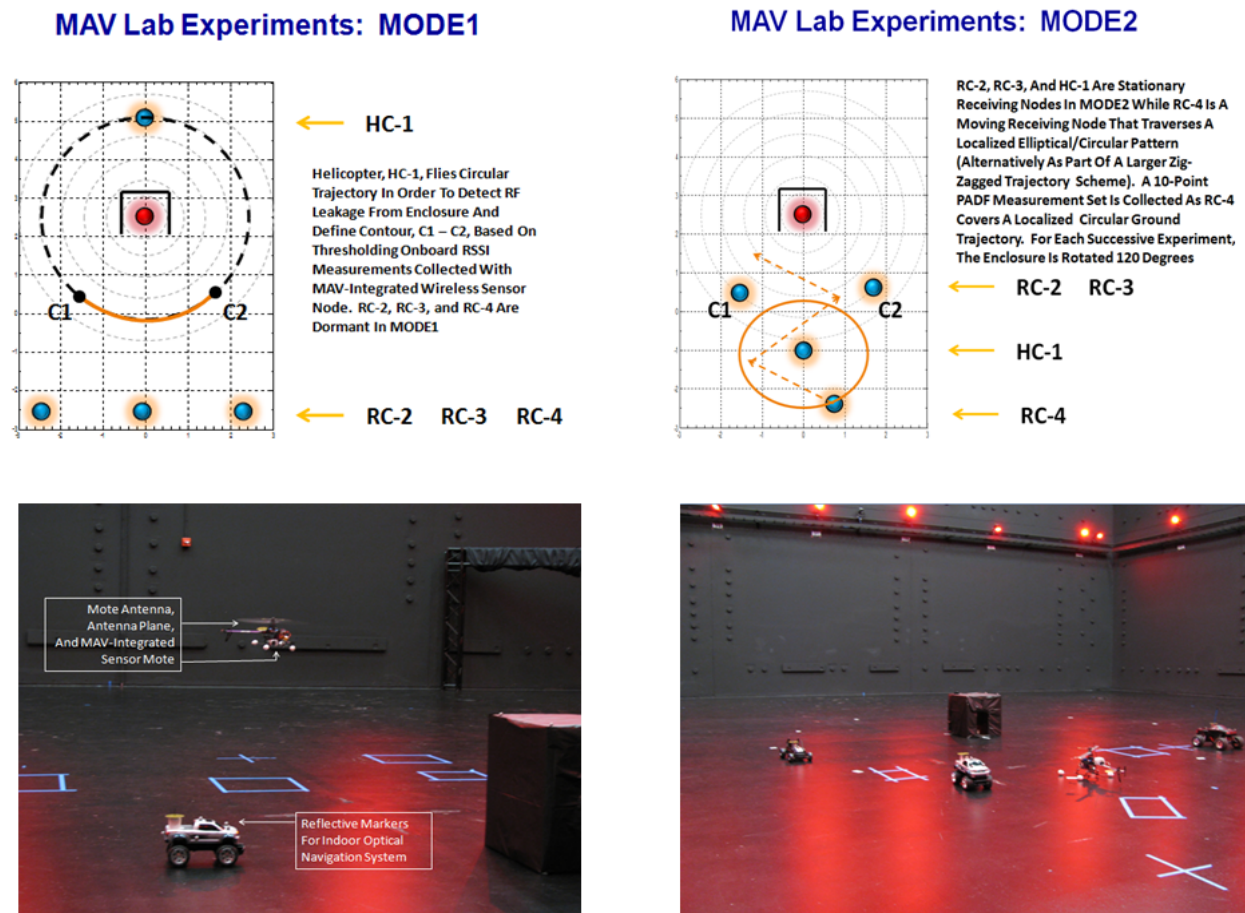


Fig. 8. Two-Mode Experimental PADF Concept

The contour limits, C1 and C2, in MODE1 are computed by applying thresholds on the RSSI data collected by the MAV during the circular trajectory encompassed by MODE1. Computation of the C1 and C2 sub-contour limits, in turn, enable the placement of two RC-controlled ground vehicles in the vicinity of points C1 and C2 during the beginning phases of MODE2 data collections. In addition, the MAV helicopter platform is placed at a slight standoff distance from the center of the detected contour interval for the RF leakage point in front of the enclosure.

During the dynamic data collection phases of the MODE2 portion of these MAV Lab Experiments, a fourth PADF sensor mote platform (i.e. another RC-controlled ground vehicle) is operated in either a localized Zig-Zag Pattern or a Localized Circular Pattern. For the data analysis in this paper, data from moving the fourth PADF sensor mote platform along a 10-point Localized Circular Pattern is presented. A complete MODE2 data set is comprised of collecting a series measurements with all four receiving platforms while operating the fourth RC platform along a 10-point Localized Circular Pattern for three different orientations of the embedded enclosure: 0 Degrees, 120 Degrees, and 240 Degrees. Two RC ground vehicles remain at stationary contour sub-interval positions C1 and C2 throughout all three sets of measurements with the three enclosure orientations; while the dynamic RC ground vehicle and the MAV helicopter

platform are re-located via a mapping function for each orientation of the enclosure.

Figure 9 – 11 are plots of the PADF Metric (Blue Line), Localization Error (Red Line), and a fluctuating RSSI-based Distance Estimate (Green Line) from the fourth dynamic ground vehicle platform to the embedded emitter.

As illustrated by the comments labeled in Orange on each of these figures, one of the most significant interpretations that can be derived via observation and analysis of these plots is the occurrence of isolated regions of relative stability associated with the Distance Parameter, D

These initial results indicate the potential for developing intelligent PADF systems that may incorporate novel and advanced processing features such as the capability to delimit, or even “Cherry Pick”, the data during dynamic PADF operation. Further pursuit of this type of PADF research shows potential for the development of PADF systems that may demonstrate a degree of robustness and localization accuracy via development and implementation of advanced distributed self-data- pruning processes. In addition, some novel platform concepts are under consideration for future tiers of research including PADF concepts with launchable projectiles and wirelessly remote-controlled integrated/ejectable RC helicopter blades and RF apertures.

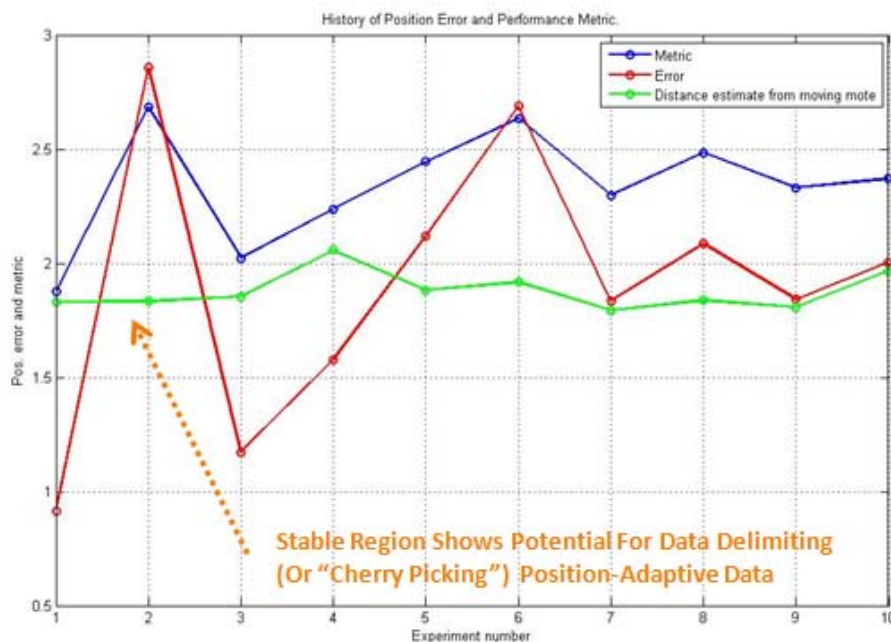


Fig. 9. MAV Lab Experiments: 0 Degree Box Rotation

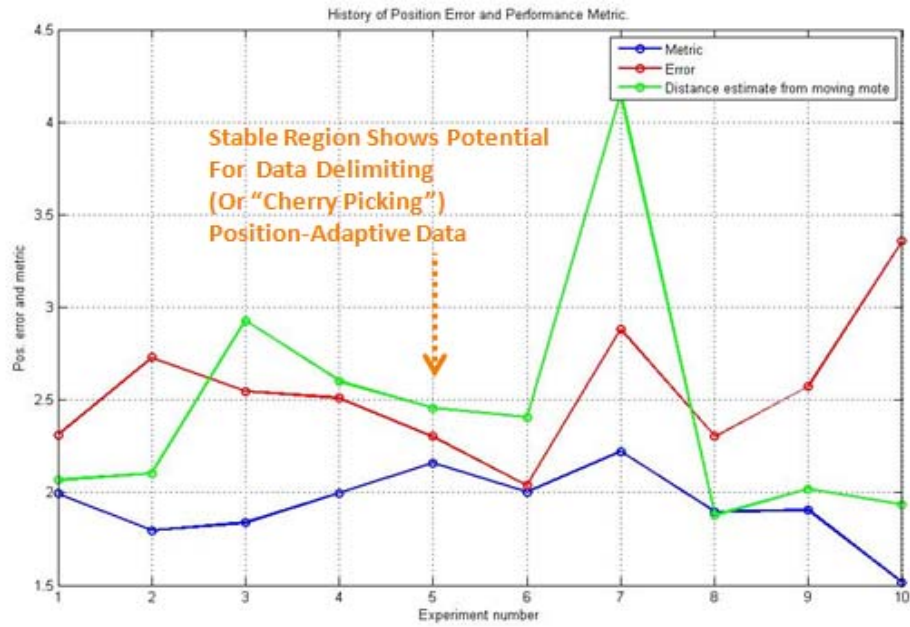


Fig. 10. MAV Lab Experiments: 120 Degree Box Rotation

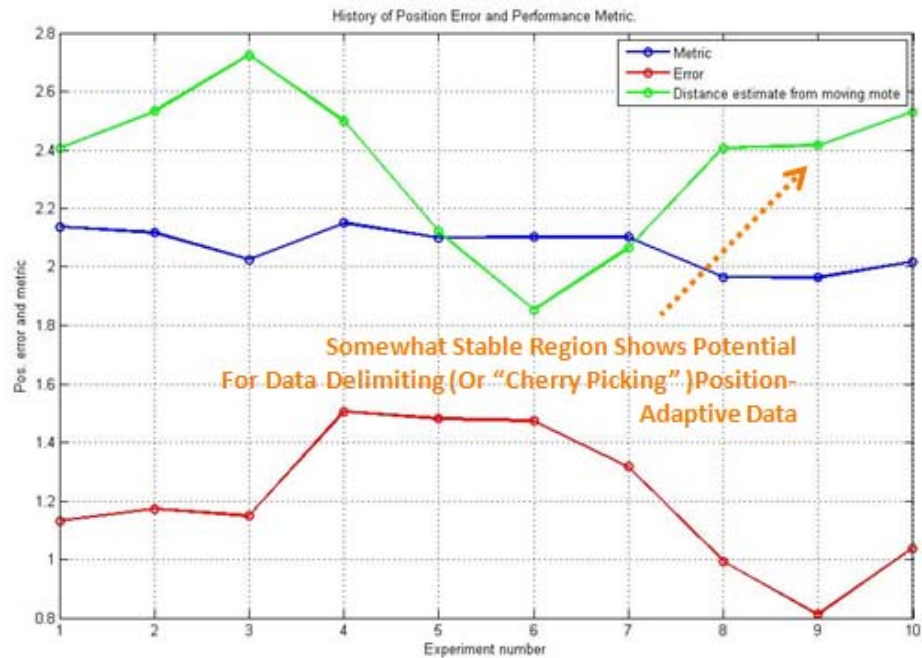


Fig. 11. MAV Lab Experiments: 240 Degree Box Rotation

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